

Python Programming Tasks

Solutions with source code and sample output

Question 1

Ask the user for Name, Age, Program and Marks, and store the information in a tuple.

Code:

```
# Question 1: Store student information in a tuple
name = input("Enter your name: ")
age = int(input("Enter your age: "))
program = input("Enter your program: ")
marks = float(input("Enter your marks: "))

student = (name, age, program, marks)

print("\nStudent information stored in tuple:")
print(student)
```

Sample output:

```
Enter your name: Enter your age: Enter your program: Enter your marks:
Student information stored in tuple:
('Radhika Bai', 20, 'BSCS', 85.5)
```

Question 2

Ask the user to enter a sentence and display the number of characters, words, vowels, spaces and digits.

Code:

```
# Question 2: Analyse a sentence
sentence = input("Enter a sentence: ")

characters = len(sentence)
words = len(sentence.split())
vowels = 0
spaces = 0
digits = 0

for ch in sentence:
    if ch.lower() in "aeiou":
        vowels += 1
    if ch == " ":
        spaces += 1
    if ch.isdigit():
        digits += 1

print("\nNumber of characters:", characters)
print("Number of words      :", words)
print("Number of vowels     :", vowels)
print("Number of spaces      :", spaces)
print("Number of digits      :", digits)
```

Sample output:

```
Enter a sentence:
Number of characters: 29
Number of words      : 7
Number of vowels     : 6
Number of spaces      : 6
Number of digits      : 5
```

Question 3

Ask the user to enter Name, Age and Salary for 3 employees. Store each employee's information in a tuple and store all tuples in a list.

Code:

```
# Question 3: Store 3 employees as tuples inside a list
employees = []

for i in range(1, 4):
    print(f"\nEnter information for employee {i}:")
    name = input("Name: ")
    age = int(input("Age: "))
    salary = float(input("Salary: "))
    employees.append((name, age, salary))

print("\nList of employees:")
print(employees)
```

Sample output:

```
Enter information for employee 1:
Name: Age: Salary:
Enter information for employee 2:
Name: Age: Salary:
Enter information for employee 3:
Name: Age: Salary:
List of employees:
[('Ali', 25, 50000.0), ('Sara', 30, 65000.0), ('Ahmed', 28, 58000.0)]
```

Question 4

Given numbers = (10, 15, 20, 25, 30, 35, 40), use a loop to count even numbers and odd numbers.

Code:

```
# Question 4: Count even and odd numbers using a loop
numbers = (10, 15, 20, 25, 30, 35, 40)

even_count = 0
odd_count = 0

for num in numbers:
    if num % 2 == 0:
        even_count += 1
    else:
        odd_count += 1

print("Even numbers:", even_count)
print("Odd numbers :", odd_count)
```

Sample output:

```
Even numbers: 4
Odd numbers : 3
```

Submitted by: **Radhika Bai**